CSC1042 Computer Network II

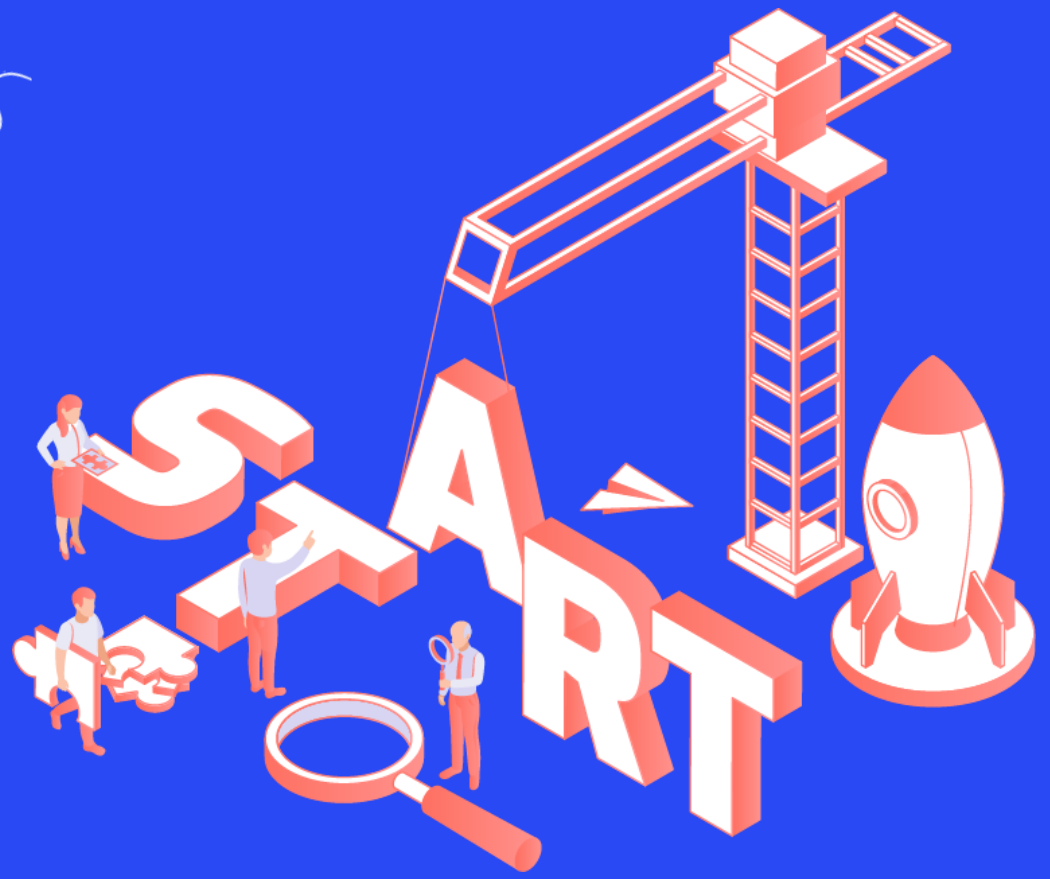
Network Address Translation

Dr. Mohammed Amine TOGOU



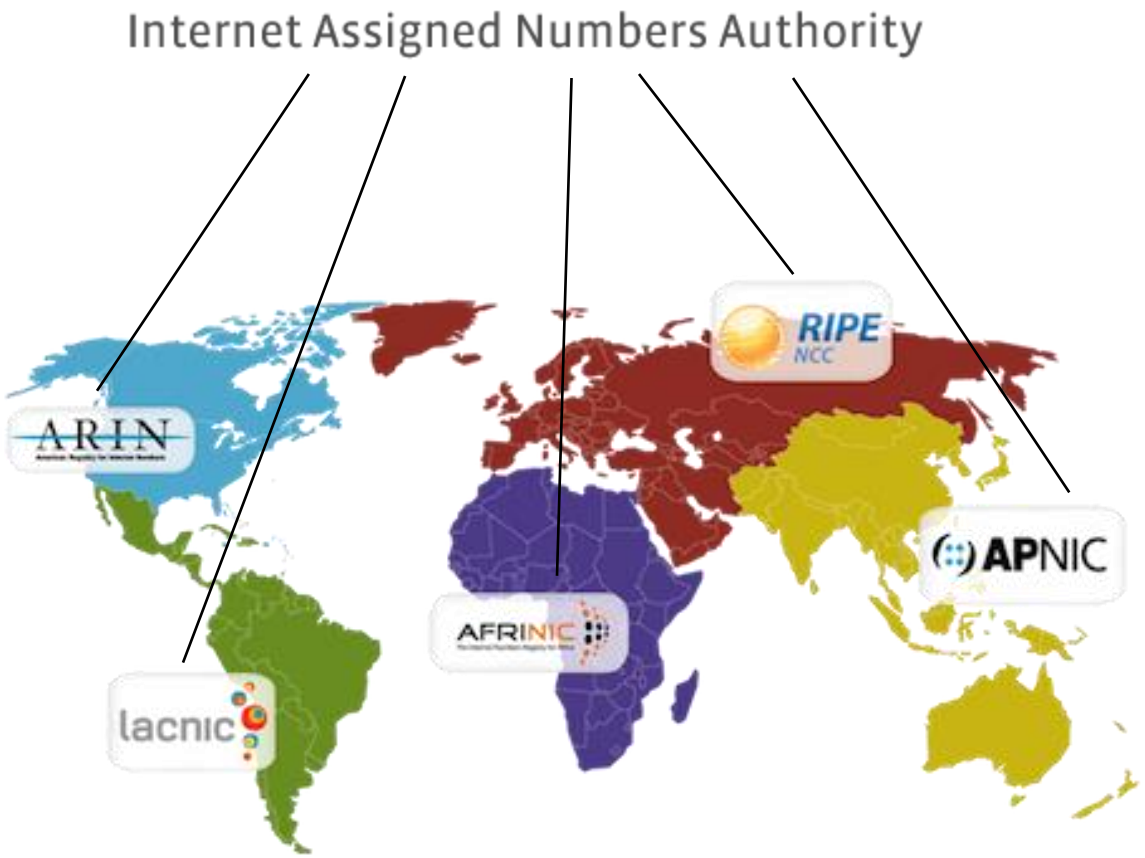
Learning outcomes

- Describe the service provided by NAT
- Explain how NAT works



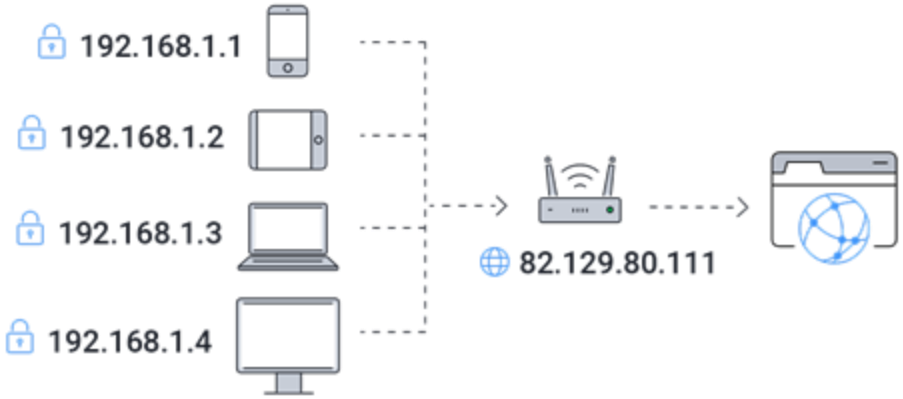
How are IP addresses assigned?

- Internet Assigned Numbers Authority (IANA) is the global coordinator of the IP addresses
- Regional Internet Registries (RIRs) are responsible for allocating IP addresses to their respective regions
- Organisations, e.g., ISPs, in these regions get IP addresses from RIRs



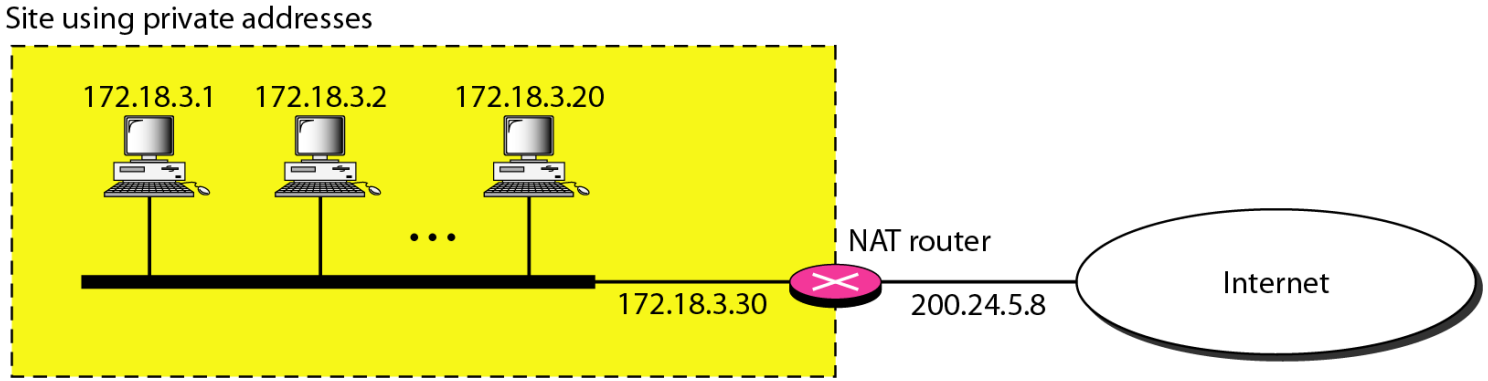
How ISPs assign IP addresses to customers?

- ISPs get blocks of IP addresses from RIRs
- ISPs assign IP addresses to their customers based on requirements
- For example, homes are usually assigned a single public IP address
  - The public IP address is assigned to the default gateway (DG)
  - Devices within the house are assigned private IP addresses by DG
  - We know that private IP addresses cannot be used to connect to the internet
  - How does the computer in the house gets access to Internet?



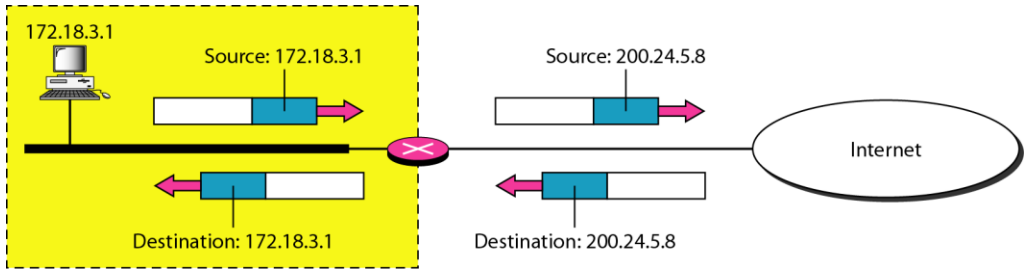
Network Address Translation - NAT

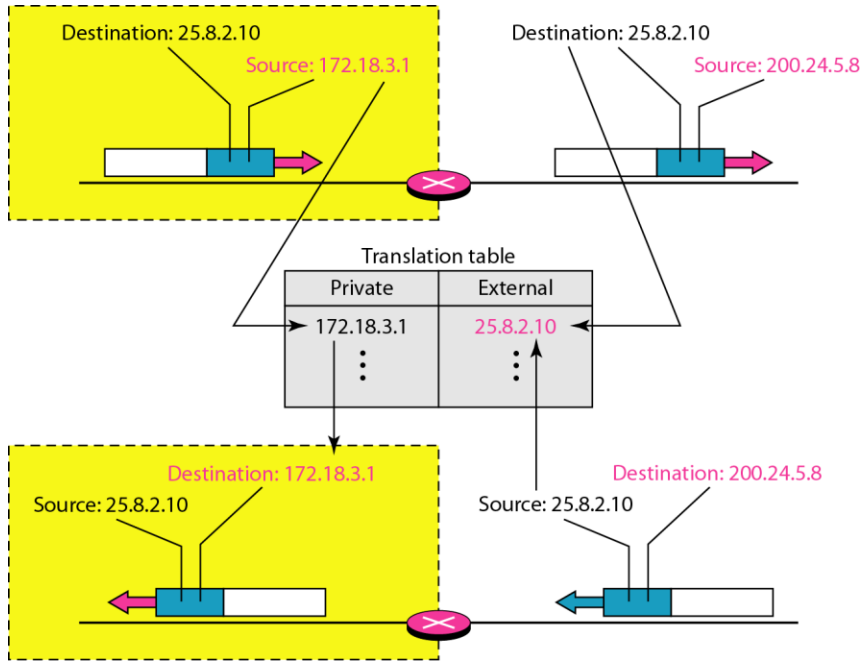
- NAT is usually implemented in routers that connect local network to the Internet
- In this figure, the router has two interfaces: one with a private address, connected to the internal network, and one with a public address, connected to the Internet
- This makes the private network transparent to the Internet as the only address shown is the NAT router address which is 200.24.5.8



Network Address Translation - NAT

- All the outgoing packets go through the NAT router
    - It replaces the source IP address in the packet with the public IP address
- All incoming packets also pass through the NAT router
    - It replaces the destination IP address in the packet with the appropriate private IP address
- It uses a translation table to decide to which private IP address the packets should be forwarded





Network Address Translation - NAT

- Assume two hosts from the private network, e.g., 172.18.3.1 and 172.18.3.2, communicate with the same destination 25.8.3.2?
- How would NAT know which packet to send to which host?
- We use a 5-column NAT translation table

Private Address	Private Port	External Address	External Port	Protocol
172.18.3.1	1400	25.8.3.2	80	TCP
172.18.3.2	1401	25.8.3.2	80	TCP
…	…	…	…	…

THANK YOU